

Assignment 3: Computational Fluid Dynamics (ME543), Aug-Nov 2024

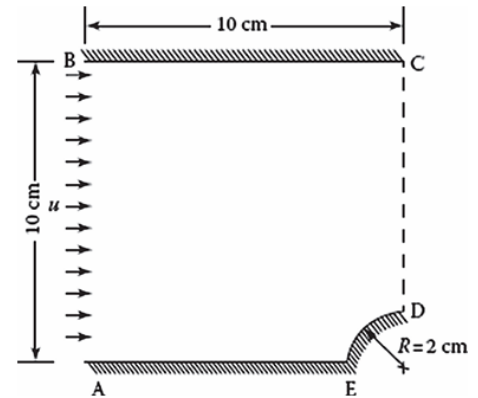
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- Provide: (i) the grid detail, (ii) the discretized equations detail, (iii) the boundary condition implementation detail, (iv) the required output (plots/any other such means), (v) a well-documented code.
- Items (i), (ii), and (iii) above should be written out/typed on a separate sheet and attached before items (iv). Soft copy of document containing items (i) to (iv) and code need to be submitted.
- Solve the following problems using finite difference method. If not provided, you can decide your own grid spacing and determine roughly the *coarsest* grid for *sufficient* accuracy, say 10^{-6} . If the analytical solution can be found for a problem then compare the numerical results with the analytical solution by plotting them together.

1. Determine the inviscid (ideal) flow in a 2-D channel containing a cylinder. Use a stream function (ψ) formulation in which the governing equation for the inviscid flow is

$$\nabla^2 \psi = 0$$

The flow domain is shown in the figure. Use a grid of $\Delta x = \Delta y = 0.5$ cm wherever possible. Unequal grid spacing will be needed near the cylinder. At the inlet (along A-B), the flow is uniform at 1 m/s. Along B-C, the stream function remains constant. Along C-D, $\partial\psi/\partial x = 0$. The stream function is zero along D-E-A.



- Determine the values of ψ throughout the flow and plot the streamline pattern for the flow.
- Determine the velocity distribution along C-D.
- From the computed results, estimate the pressure coefficient at the top of the cylinder (point D) and compare this with the pressure coefficient from the “exact” analytical solution for inviscid flow around a cylinder in a stream of infinite extent.

2. Solve the linearized Burgers’ equation using (a) FTCS and (b) MacCormack schemes. For the initial conditions $u(x, 0) = 0$ for $0 \leq x \leq 1$ and boundary conditions $u(0, t) = 100$, $u(1, t) = 0$. Use 21 grid point mesh. Find the steady-state solution for $r = 0.10$ and $C_0 = 0.40$, and compare the numerical solutions with the exact solution.

3. Solve the generalized Burgers’ equation

$$u_t + \left(\frac{1}{2} - u\right)u_x = 0.001u_{xx}$$

using MacCormack scheme for the initial condition

$$u = \frac{1}{2}\{1 + \tanh[250(x - 20)]\}, \quad 0 \leq x \leq 40$$

and exact Dirichlet boundary conditions. Choose a 41 grid point mesh with $\Delta x = 1$ and compute to $t = 18$. Solve this problem for $\Delta t = 1.0$ and 0.5 , and compare graphically with the exact stationary solution.

Coding guidelines/tips

- Refer assignment 1.